

Waveform Coding

Prediction Filter:

Linear Prediction:

Prediction is used for estimating the future samples of a signal from present and past input sample values.

The prediction is said to be linear if the future sample value is a linear combination of present and past input sample values.

The predicted sample value is given by

$$\hat{x}(n) = \sum_{k=1}^M W_k x(n-k) \rightarrow \textcircled{1}$$

where $\hat{x}(n)$ is predicted future sample

$x(n-k) \rightarrow$ past sample

$\hookrightarrow x(n-1), x(n-2), \dots, x(n-k)$

$W_k \rightarrow$ set of multiplier co. eff.

Prediction error:

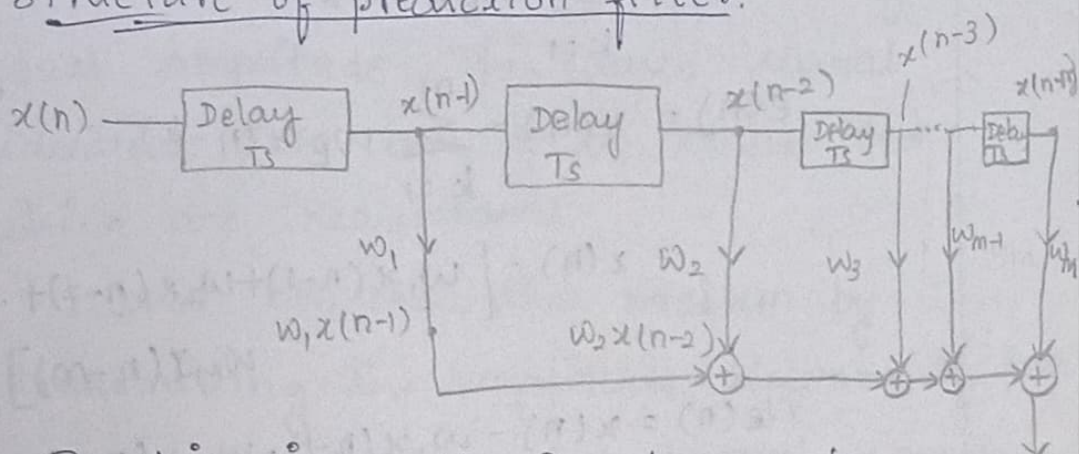
The error that happens because of prediction is called prediction error. Prediction error is the difference b/t the actual sample value and the predicted sample value.

Prediction filter:

Implementation of linear predictor equation

$\hat{x}(n) = \sum_{k=1}^m w_k x(n-k)$ is known as prediction filter.

Structure of prediction filter:



In this figure, the input sample $x(n)$ is delayed by one sampling duration T_s . Hence we get $x(n-1)$ which is the previous input sample. Similarly, if $x(n-1)$ is delayed by T_s , we get $x(n-2)$ and so on. The past input samples are multiplied by multiplier coefficients $w_1, w_2, \dots, w_m$ and are added to get $\hat{x}(n)$. This structure is called a filter because eqn ① represents linear convolution and such convolution represents filtering. In this structure $x(n-1), x(n-2), \dots, x(n-m)$ are the past inputs which are used for filtering with the help of multiplier coefficients $w_1, w_2, \dots, w_m$. This filter is of the type FIR digital filter. The filtering operation can be modified by changing the filtering coefficients.

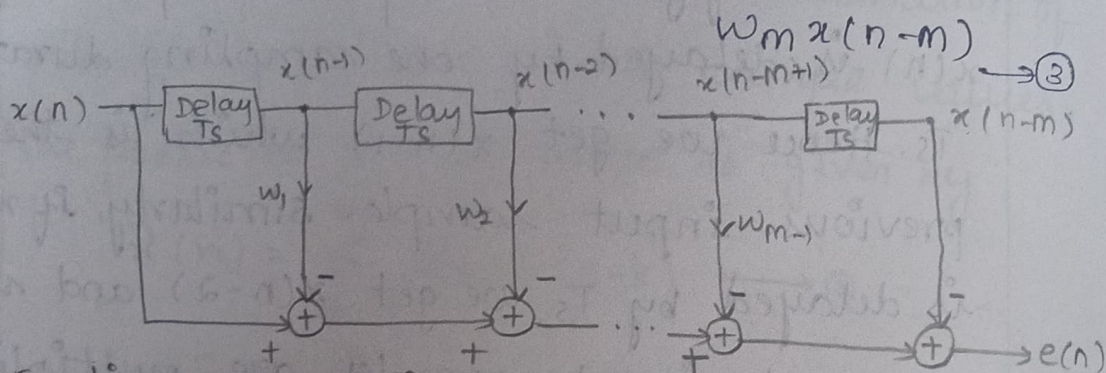
structure of prediction error:

The prediction error $e(n) = x(n) - \hat{x}(n)$ where $\hat{x}(n) = \sum_{k=1}^m w_k x(n-k)$ → ②

$$e(n) = x(n) - \sum_{k=1}^m w_k x(n-k)$$

$$= x(n) - [w_1 x(n-1) + w_2 x(n-2) + \dots + w_m x(n-m)]$$

$$e(n) = x(n) - w_1 x(n-1) - w_2 x(n-2) - \dots -$$



For linear predictive coding (write up to this).

Application of prediction filter:

⇒ Prediction filter is used in DPCM system to minimize the error applied to the DPCM quantizer.

DPCM - Differential Pulse Code modulation.

Need for DPCM [or Limitation of PCM]:

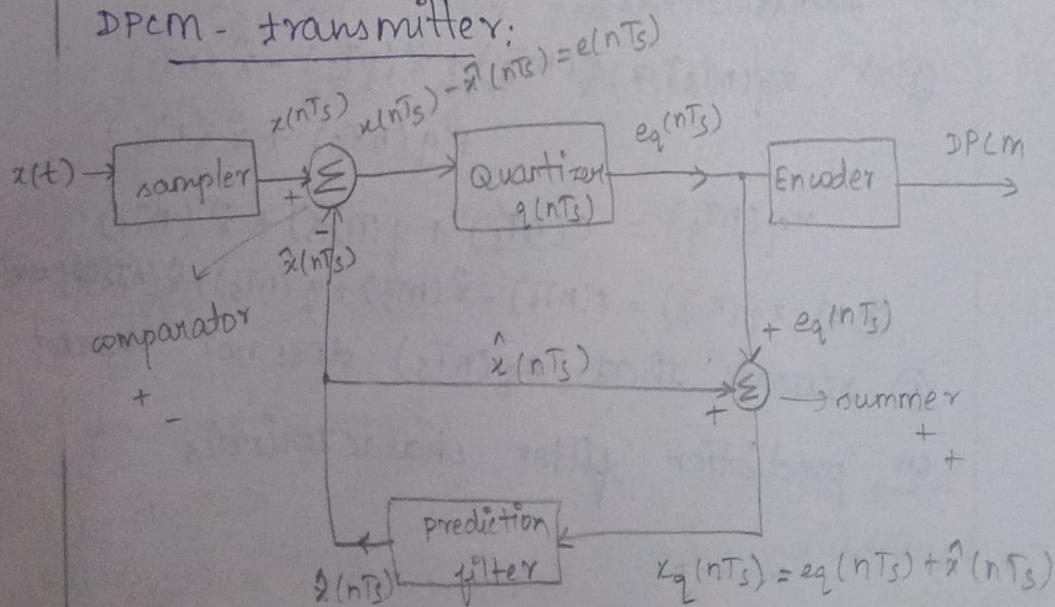
1. In PCM, the adjacent samples are of almost equal amplitude. This causes several identical PCM codes to be transmitted which are redundant.

2. DPCM overcomes this problem by transmitting the amplitude difference of two successive samples instead of actual samples. Hence fewer bits are required for DPCM.

Principle of DPCM:

In DPCM, redundancy and overall bit rate will decrease. No. of bits required to transmit one sample will also be reduced by transmitting only the difference between consecutive samples.

DPCM - transmitter:



The input signal $x(t)$ is given to the sampler. The output of sampler $x(nT_s)$ and a predicted sample $\hat{x}(nT_s)$ is fed to comparator. The output of comparator $e(nT_s)$ is sent to the quantiser where quantisation error gets added. The output of quantiser $e_q(nT_s)$ is fed to the encoder. The output of quantiser and the prediction filter's output are given to a summer and its output $x_q(nT_s)$ is passed to the prediction filter.

$$x_q(nT_s) = e_q(nT_s) + \hat{x}(nT_s) \rightarrow (1)$$

$$e_q(nT_s) = e(nT_s) + q(nT_s) \rightarrow (2)$$

where $q(nT_s)$ is the quantisation error.

$$(1) \Rightarrow x_q(nT_s) = e(nT_s) + q(nT_s) + \hat{x}(nT_s)$$

$$e(nT_s) = x(nT_s) - \hat{x}(nT_s)$$

$$x_q(nT_s) = x(nT_s) - \hat{x}(nT_s) + q(nT_s) + \hat{x}(nT_s) \rightarrow (3)$$

$$x(nT_s) = e(nT_s) + \hat{x}(nT_s) \rightarrow (4) \text{ from sampler.}$$

sub (4) in (3)

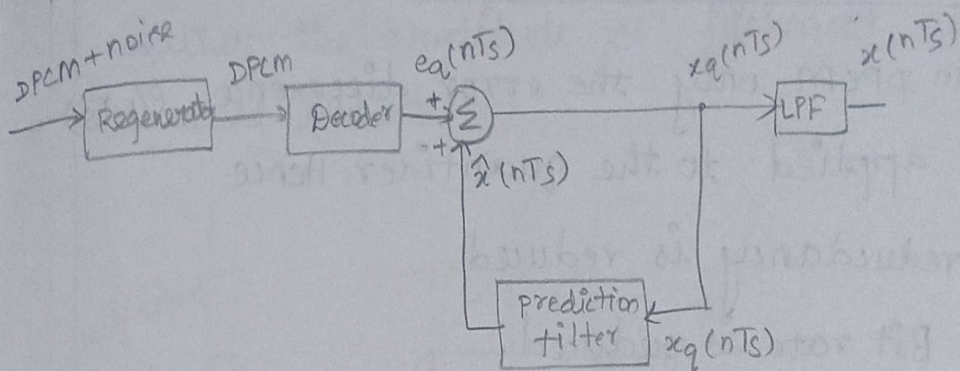
$$(3) \Rightarrow x_q(nT_s) = e(nT_s) + \hat{x}(nT_s) - \hat{x}(nT_s) + q(nT_s) + \hat{x}(nT_s)$$

$$x_q(nT_s) = e(nT_s) + q(nT_s) + \hat{x}(nT_s)$$

$$(3) \Rightarrow x_q(nT_s) = x(nT_s) - \hat{x}(nT_s) + q(nT_s) + \hat{x}(nT_s) = x(nT_s)$$

(5) reveals that $x_q(nT_s)$ does not depend on prediction filter characteristics

DPCM Receiver:



⇒ DPCM Rx comprises of regenerator, decoder, summer, prediction filter, LPF blocks.

⇒ Regenerator block receives the DPCM signal & noise signal that fed into decoder.

⇒ In decoder, it decodes the signal and produces the signal $e_q(nT_s)$.

⇒ In summer, it receives two signals one is $e_q(nT_s)$ and o/p of prediction filter $\hat{x}(nT_s)$.

⇒ Then the o/p of summer gives $x_q(nT_s)$ that move to the LPF which is also a smoothening filter and smoothen the signal and give $x(nT_s)$ i.e msg signal.

$s) + q(nT_s)$

→ ⑤

redundant information:

In PCM

If three successive samples have same amplitude, it is sent as it is

In DPCM

redundancy ↓

diff. of amplitude is sent.

redundant - nearly same info. is repeated
Advantages of DPCM over PCM:

⇒ In DPCM, only the error difference $e(nT_s)$ is applied to the quantiser. Hence

1. redundancy is reduced.
2. Bit rate is reduced.
3. Transmission BW is reduced.

Disadvantages:

1. In DPCM, the system complexity is increased because of the use of prediction filter.

Delta Modulation:

In DPCM $\rightarrow x(nT_s) - \hat{x}(nT_s) = e(nT_s) \rightarrow \text{encoded} \rightarrow T_x$.

In DM $\rightarrow x(nT_s) - \hat{x}(nT_s)$
 $\downarrow$
 $e(nT_s)$
 $\downarrow$
error

$\left. \begin{array}{l} +ve = +1 \text{ Step size} \\ +8 \rightarrow 1 \\ -ve = -8 \rightarrow 0 \end{array} \right\} \text{encoding.}$

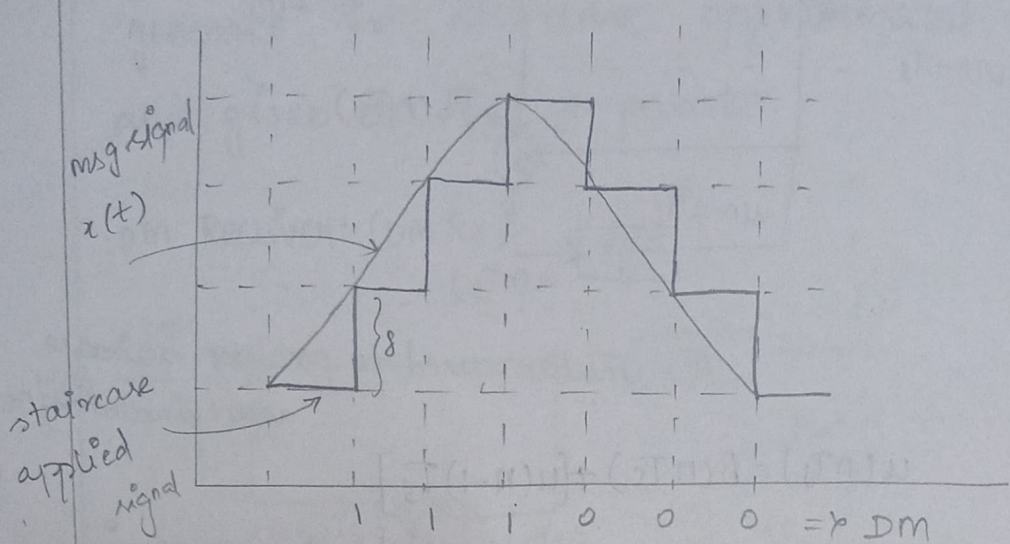
Comparison of PCM and DM:

1. PCM transmits all bits which are used to code the sample. Hence, the signaling rate and transmission BW are large in PCM. To overcome this problem DM is used which transmits only one bit per sample.

Operating Principle of DM:

DM transmits only one bit per sample
i.e., the present sample value is compared

with previous sample value and the indication whether the amplitude is increased or decreased is sent.



$$+ \delta \left\{ \begin{array}{l} \text{if } x(t) > \text{staircase} \\ \text{then } +1 \end{array} \right. \quad - \delta \left\{ \begin{array}{l} \text{if } x(t) < \text{staircase} \\ \text{then } -1 \end{array} \right. = 0$$

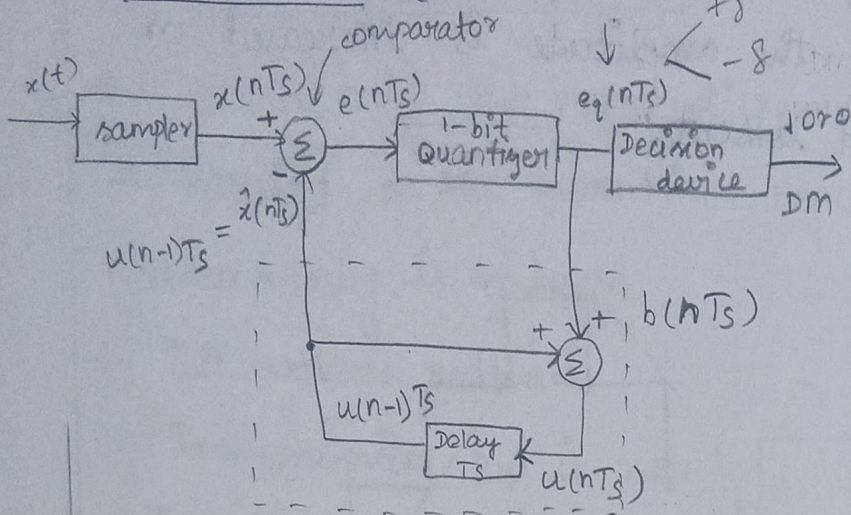
The input signal $x(t)$ is approximated to step signal (staircase approximated signal) by delta modulator. The step size is fixed. The difference between the input signal $x(t)$ and the staircase approximated signal is confined to two levels. ($+\delta$ and $-\delta$)

If the difference is positive, then the approximated signal is increased by one step size ($+\delta$) and bit '1' is transmitted.

If the difference is negative, then the approximated signal is decreased by one step size ($-\delta$) and bit '0' is transmitted.

DM Transmitter (or)

Delta Modulator.



Accumulator - used to produce staircase approximated signal.

$$u(nT_s) = b(nT_s) + [u(n-1)T_s]$$

1-bit-Quantizer is used to find the error is +ve or -ve.

Accumulator consists of Delay, summer.

⇒ The first block of Delta modulator is sampler which is used to convert continuous signal $x(t)$ to discrete signal $x(nT_s)$.

⇒ Then $x(nT_s)$ is sent to comparator, and $\hat{x}(nT_s)$ is also sent to comparator and produces the output $e(nT_s)$ i.e., error signal.

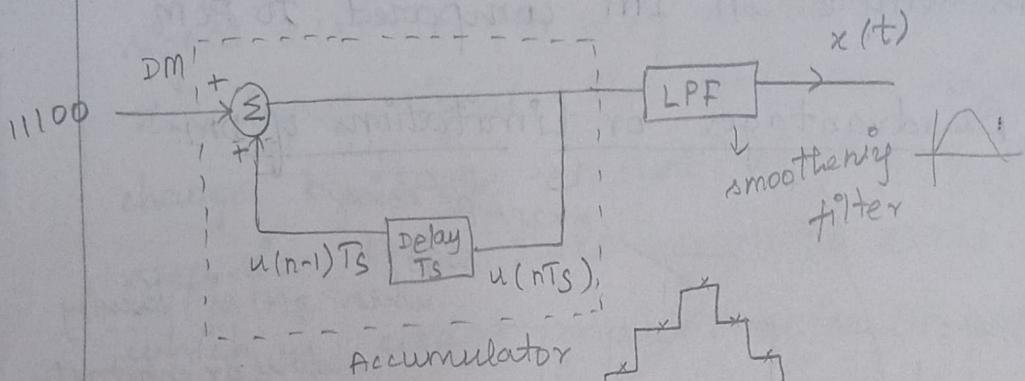
⇒ $e(nT_s)$ is given to 1-bit quantiser and gives $e_q(nT_s)$ or $b(nT_s)$.

⇒ $b(nT_s)$ is generated based on the o/p of comparator $e(nT_s)$; if $e(nT_s)$ is +ve then o/p of quantizer +8 or else produces -8.

⇒ This will fed to decision device whether it transmits 1 or 0 based on o/p of

quantiser. In feedback path, there is an accumulator which consists of summer, delay unit. With the help of accumulator, we generate the staircase approximated signal and given to the comparator.

DM Receiver: (DM Rx)



Use regenerator before summer, which will remove noise.

At o/p of DM Rx, 100% reconstruction does not happen due to "quantiser".

⇒ The DM signal and the delayed signal $u(n-1)Ts$ are fed as inputs to a summer.

⇒ The o/p of summer is given to a LPF which is actually a smoothing filter.

⇒ At the output of LPF, we get back the signal $x(t)$.

⇒ The summer and the delay unit altogether acts as an accumulator.

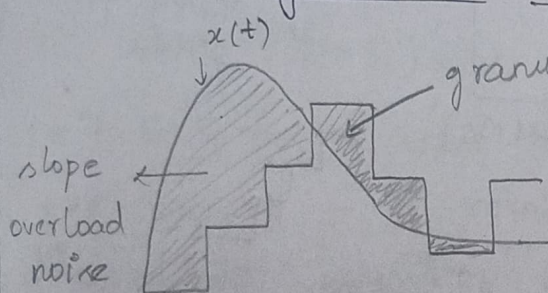
Advantages of DM over PCM: bit rate - no. of bits transmitted per sample

⇒ The bit rate and transmission BW is reduced in DM as it transmits only 1 bit per sample.

⇒ The transmitter and receiver implementation is simple in DM compared to PCM.

① Disadvantages or Limitations of DM:

8m



when $x(t)$ is varying slowly or constant.

⇒ δ is fixed.

⇒ To overcome,

$\delta \rightarrow \downarrow$

when $x(t)$ is steep

⇒ δ is fixed.

⇒ To overcome, $\delta \rightarrow \uparrow$

Slope overload distortion (or) noise:

This distortion arises because of large dynamic range of the input signal.

Referring the diagram shown, the rate of rise of input signal $x(t)$ is so high that the staircase signal cannot approximate since the step size δ becomes too small for the staircase signal to follow the steep segment of $x(t)$. Thus there exists an error or distortion which is called as

slope overload distortion. To reduce this error, step size should be $\uparrow$ when slope of the signal $x(t)$ is high.

Granular noise:

It occurs when the step size is too large compared to small variation in the signal $x(t)$ i.e., for very small variations in the input signal, the staircase signal changes by large amount because of fixed step size. Hence there occurs an error which is called granular noise. To avoid granular noise, step size should be made small.

1. Consider a sine wave of freq. f_m and amplitude A_m is applied to a Delta modulator with step size s . Show that slope overload distortion will occur if $A_m > \frac{s}{2\pi f_m T_s}$ where T_s is the sampling period of DM.

soln:

Let us assume a sinusoidal wave of freq. f_m and amplitude A_m .

$$x(t) = A_m \sin(2\pi f_m t)$$

$$\text{slope of sine wave } x(t) = \left| \frac{dx(t)}{dt} \right|$$

$$\text{slope of DM} = \frac{\text{step size}}{\text{sampling period}} = \frac{s}{T_s}$$

slope overload noise will occur if

slope of $x(t)$ > slope of DM

$$\left| \frac{dx(t)}{dt} \right| > \frac{\delta}{T_s}$$

$$\left| \frac{d(A_m \sin(2\pi f_m t))}{dt} \right| > \frac{\delta}{T_s}$$

$$|A_m \cos(2\pi f_m t) 2\pi f_m| > \frac{\delta}{T_s}$$

$$2\pi A_m f_m |\cos(2\pi f_m t)| > \frac{\delta}{T_s}$$

$$\cos \theta_{\max} = 1$$

$$2\pi A_m f_m (1) > \frac{\delta}{T_s}$$

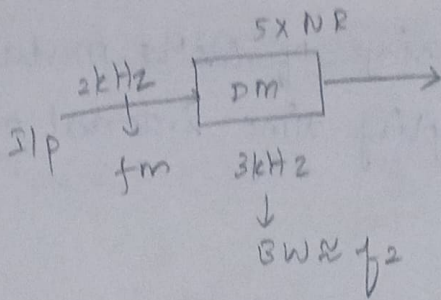
$$A_m > \frac{\delta}{2\pi f_m T_s}$$

To overcome slope overload noise,

$$A_m < \frac{\delta}{2\pi f_m T_s}$$

2. A delta modulator is designed to operate at 5 times Nyquist rate for a signal with 3 kHz bandwidth. Determine the maximum amplitude of a 2 kHz input sinusoid for which the delta modulator does not have slope overload. Quantisation step size is 200 mV. Derive the formula that you use.

soln:



For DM,

$$f_s = 5 \times \text{Nyquist rate}$$

$$= 5 \times 2 f_2$$

$$f_s = 10 f_2 = 10 \times 3 \text{ kHz}$$

$$f_s = 30 \text{ kHz}$$

$$T_s = \frac{1}{30 \text{ k}} = 0.0333 \text{ ms}$$

$$T_s = 33.3 \mu\text{s}$$

slope overload will not occur if

$$A_m < \frac{\delta}{2\pi f_m T_s} \quad \left| \frac{dx(t)}{dt} \right| < \frac{\delta}{T_s}$$

$$< \frac{200 \times 10^{-3}}{2\pi \times 2 \times 10^3 \times 0.0333 \times 10^{-3}}$$

$$< 478.185 \times 10^{-3}$$

$$A_m < 0.478 \text{ V}$$

$$A_m < 478 \text{ mV}$$

⑧ Signal to Quantisation Noise Ratio of DM:

Assume that there is no slope overload noise. $A_m < \frac{\delta}{2\pi f_m T_s} \rightarrow \text{①}$

where $A_m \rightarrow$ Amplitude of the input (msg) signal
 $f_m \rightarrow$ max. freq. of input signal.

$\delta \rightarrow$ step size of Delta modulator

$T_s \rightarrow$ sampling time interval of Delta modulator

signal power:

$$P = \frac{V_{\text{signal}}^2}{R} \rightarrow (2)$$

where $V_{\text{signal}}^2 \rightarrow$ mean square value of input signal

Assume input as sinusoidal signal which is normalized to 1.

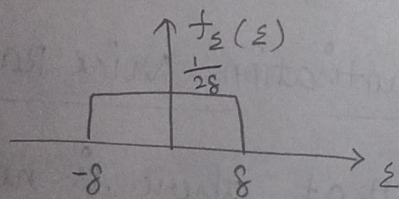
$$P = \left(\frac{A_m}{\sqrt{2}} \right)^2 \rightarrow (3)$$

Sub (1) in (3).

$$P = \frac{\left(\frac{\delta}{2\pi f_m T_s} \right)^2}{2} = \frac{\delta^2}{8\pi^2 f_m^2 T_s^2} \rightarrow (4)$$

Noise power:

Maximum quantisation error in DM is equal to the step size δ . Let the quantisation error, we uniformly distributed over the interval $(-\delta, +\delta)$.



$$\frac{1}{b-a} = \frac{1}{8-(-8)} = \frac{1}{28}$$

$$f_z(z) = \begin{cases} \frac{1}{28} & ; -8 \leq z \leq 8 \\ 0 & ; \text{otherwise} \end{cases}$$

$$\text{Noise power} = \frac{V_{\text{noise}}^2}{R} \rightarrow (5)$$

where V_{noise}^2 is mean square value of noise.

$$V_{\text{noise}}^2 = E(x^2) = \int_{-\infty}^{\infty} \xi^2 f_{\xi}(\xi) d\xi$$

$$= \int_{-\delta}^{\delta} \xi^2 \frac{1}{2\delta} d\xi$$

$$= \frac{1}{2\delta} \int_{-\delta}^{\delta} \xi^2 d\xi$$

$$= \frac{1}{2\delta} \left[\frac{\xi^3}{3} \right]_{-\delta}^{\delta}$$

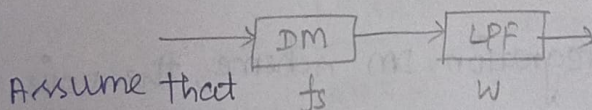
$$= \frac{1}{2\delta} \left[\frac{\delta^3}{3} + \frac{\delta^3}{3} \right]$$

$$= \frac{\delta^2}{3}$$

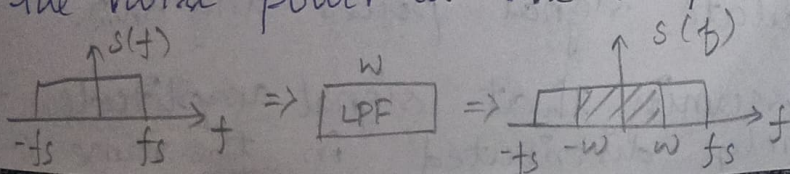
$$V_{\text{noise}}^2 = \frac{\delta^2}{3} \rightarrow (6)$$

with $R=1$ (normalised),

$$(6) \Rightarrow \text{Noise power} = V_{\text{noise}}^2 = \frac{\delta^2}{3} \rightarrow (7)$$



The noise power is uniformly distributed over $-f_s$ to $+f_s$ range. At the output of DM receiver, there is a low pass reconstruction filter whose cut-off freq. is ' w '. This cutoff freq. is equal to highest signal freq. The reconstruction filter passes part of the noise power at the output.



$$\text{output noise power} = \frac{W}{f_s} \quad \text{noise power} = \frac{W}{f_s} \times \frac{8}{3}$$

SNR (post-filtered):

$$\text{SNR} = \frac{\text{Signal power}}{\text{Noise power (post filtered)}}$$

$$= \frac{8^2}{8\pi^2 f_m^2 T_s^2}$$

$$\frac{W}{f_s} \times \frac{8^2}{3}$$

$$= \frac{3 f_s}{8\pi^2 f_m^2 T_s^2 W}$$

$$\boxed{\text{SNR} = \frac{3}{8\pi^2 f_m^2 T_s^3 W}}; \quad \left[f_s = \frac{1}{T_s} \right]$$

[Considering only the granular noise and post filtered].

1. In a single integration DM scheme, the voice signal is sampled at the rate of 64 kHz. The maximum signal amplitude is 1 V and signal BW is 3.5 kHz.
 - i) Determine the minimum value of step size to avoid slope-overload.
 - ii) Determine the granular noise N_0
 - iii) Assume the signal to be sinusoidal and calculate signal power and SNR.

in assuming that signal amplitude is uniformly distributed in the range $(-1, 1)$.

Determine signal power and amplitude.

soln:

Given,

i) $A_m = 1V$, $f_m = f_2 = BW = 3.5kHz$, $f_s = 64kHz$

$$T_s = \frac{1}{f_s} = 0.0156ms = 15.6\mu s$$

$$A_m \leq \frac{8}{2\pi f_m T_s}$$

$$A_m \cdot 2\pi f_m T_s \leq 8$$

$$1 \times 2 \times 3.14 \times 3.5 \times 10^3 \times 15.6 \times 10^{-6} \leq 8$$

$$0.342 \leq 8$$

$$8 \geq 0.342$$

$$8 = 0.34V \checkmark \quad \text{or } 8 = 342mV.$$

ii) Granular noise power: N_0

$$N_0 = \frac{W}{f_s} \times \frac{\delta^2}{3}$$

$$W = f_m$$

$$= \frac{3.5 \times 10^3}{64 \times 10^3} \times \frac{(0.34)^2}{3}$$

$$N_0 = 2.107 \times 10^{-3}$$

$$N_0 = 2.107mW$$

iii)

$$P_{\text{signal}} = \frac{\delta^2}{8\pi^2 f_m^2 T_s^2} = \frac{(0.34)^2}{8\pi^2 (3.5 \times 10^3)^2 (15.6 \times 10^{-6})^2}$$

$$= 0.4916 \approx 0.5W \checkmark$$

$$P_{\text{signal}} = \left(\frac{A_m}{\sqrt{2}} \right)^2 = \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2} = 0.5W \checkmark$$

$$SNR = \frac{P_{\text{sig}}}{P_{\text{noise}}} = \frac{3}{8\pi^2 f_m^2 T_s^3 W} \quad \text{428}$$

$$SNR =$$

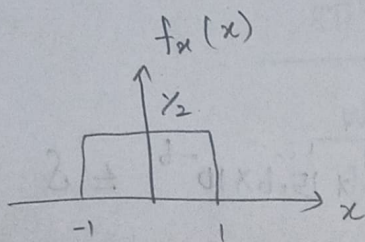
$$8\pi^2 \times (3.5 \times 10^3)^2 \times (15.6 \times 10^{-6})^3 \times 3.5 \times 10^3$$

$$= 233.665$$

SNR $\approx 234 \rightarrow$ This is post-filtered SNR

$$SNR = \frac{P_{sig}}{P_{noise}} = \frac{0.5}{0.002107} = 237.3 \approx 237$$

iv)



$$f_x(x) = \begin{cases} \frac{1}{2} & -1 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$P = \frac{V^2 \text{signal}}{R} \quad [R=1]$$

$$P = V^2 \text{signal} = E[x^2] = \int_{-1}^1 x^2 f_x(x) dx$$

$$= \int_{-1}^1 x^2 \frac{1}{2} dx = \frac{1}{2} \left[\frac{x^3}{3} \right]_{-1}^1$$

$$= \frac{1}{2} \left[\frac{1}{3} + \frac{1}{3} \right] = \frac{1}{2} \left[\frac{2}{3} \right] = \frac{1}{3}$$

$$P_{sig} = 0.3333 \text{ W} \quad \checkmark$$

$$A_{sig} = \frac{1}{2} V \quad \checkmark$$

$$SNR = \frac{P_{sig}}{P_{noise}} = \frac{0.3333}{0.002107} \approx 158.18$$

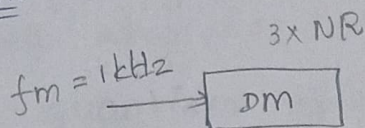
2. A DM system is designed to operate at three times the Nyquist rate for a signal

with 3kHz BW. The quantisation step size is 250mV.

i) Determine maximum amplitude of 1kHz input sinusoid for which DM doesn't show slope overload.

ii) Determine the post-filtered output SNR for the signal of part i).

soln:



$$BW \approx f_2 = 3\text{kHz}$$

$$i) f_s = 3 \times NR = 3 \times 2f_2 = 6f_2 = 6 \times 3\text{kHz} = 18\text{kHz}$$

$$T_s = \frac{1}{f_s} = 0.0556\text{ms} = 55.6\mu\text{s}$$

$$A_m < \frac{\delta}{2\pi f_m T_s}$$

$$< \frac{250 \times 10^{-3}}{2\pi \times 1 \times 10^3 \times 55.6 \times 10^{-6}}$$

$$A_m < 7.1087 \times 0.7159$$

$\therefore$ max. amplitude = 0.7 V

$$ii) SNR = \frac{3}{8\pi^2 f_m^2 T_s^3 W}$$

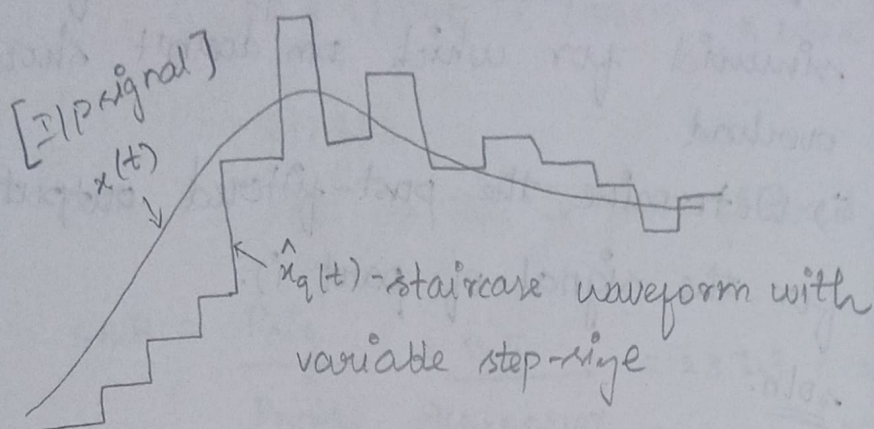
Here $W = 3\text{kHz}$

$$= \frac{3}{8\pi^2 (1 \times 10^3)^2 (55.6 \times 10^{-6})^3 (3 \times 10^3)}$$

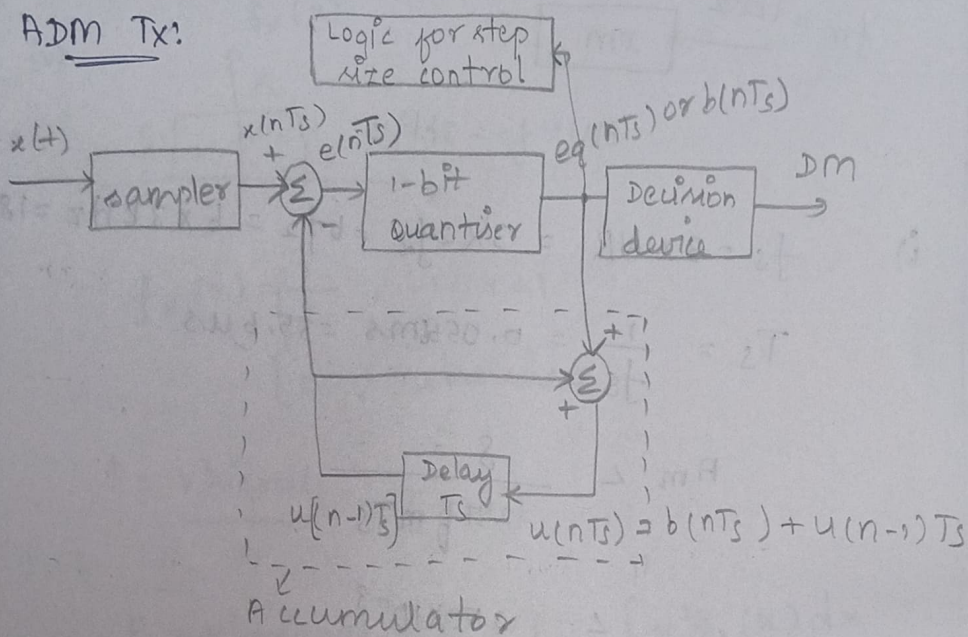
$$= 216574.18 \times 22 \times 10^8$$

$$SNR = 73.76 \approx 74 \checkmark$$

ADM - Adaptive Delta Modulator:



ADM Tx:



⇒ The output of 1-bit quantiser is given to the logic for step size control. — explain in block diagram

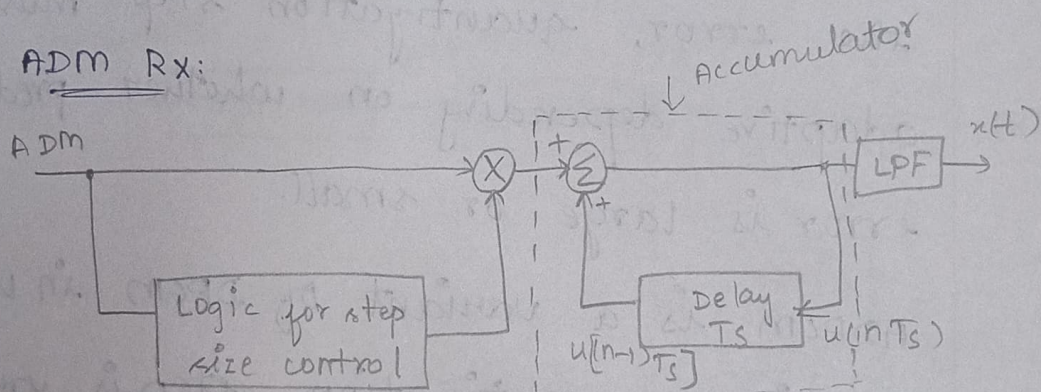
To overcome the quantisation error due to slope overload and granular noise, the step size in ADM is made adaptive to the variation in the input signal $x(t)$. In the steep segment of input signal $x(t)$, the step size is increased while when the input is varying slowly, the step size is reduced. ADM can take continuous changes or

discrete changes in step size. In Tx section, the logic for step size control is added. The step size is increased or decreased according to certain rules depending on 1-bit quantiser output.

For eg., if 1-bit quantiser op is 1 consecutively, then step size may be doubled for next sample. If quantiser op is 0 consecutively, then step size may be decreased.

Block diagram explain same as DM.

ADM Rx:



In ADM Rx, the first part generates the step size from each incoming bit. Exactly the same process is followed for step size control as there that in Tx. The previous i/p and present i/p decides the step size.

Then the output is given to accumulator which will generate staircase approximated waveform. Then it is fed to the LPF (smoothing filter) to get the required signal $x(t)$.

ADPCM - Adaptive Differential Pulse Code

Modulation:

=> used in speech coding

=> quantiser is non-uniform quantiser used as adaptive quantiser.

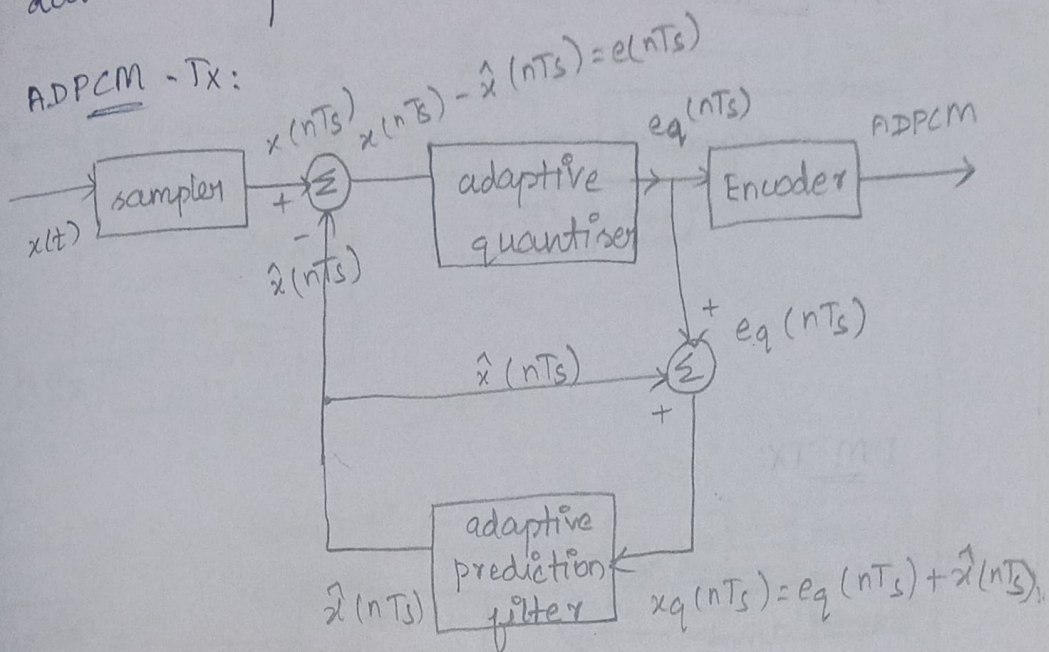
Need for ADPCM:

No. of quantization levels q in DPCM is fixed. Quantisation error occurs due to the fixed quantization step. To avoid the error, quantization step must be adaptive depending on whether prediction error is large or small.

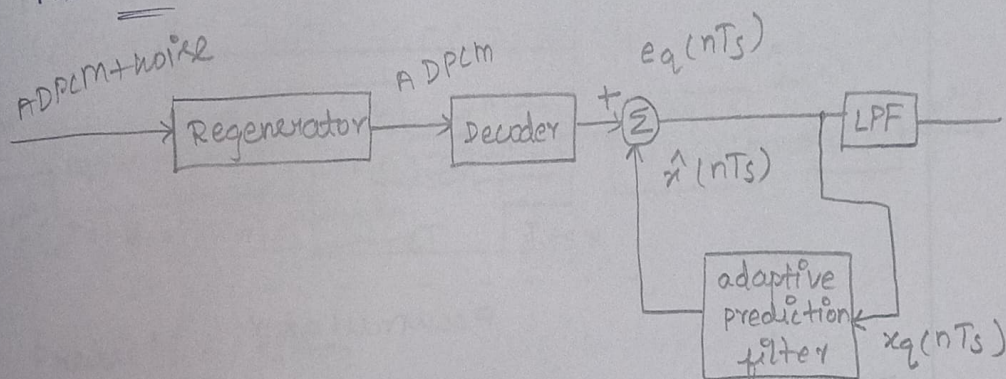
ADPCM is a variant of DPCM in which the size of quantization step is varied to allow further reduction of the required data bandwidth for the given SNR. ADPCM improves the efficiency of DPCM by incorporating adaptive quantizer at transmitter. ADPCM also uses adaptive predictor which adjusts according to the value of each input sample and thereby reducing the no. of bits required to represent the

data sample.

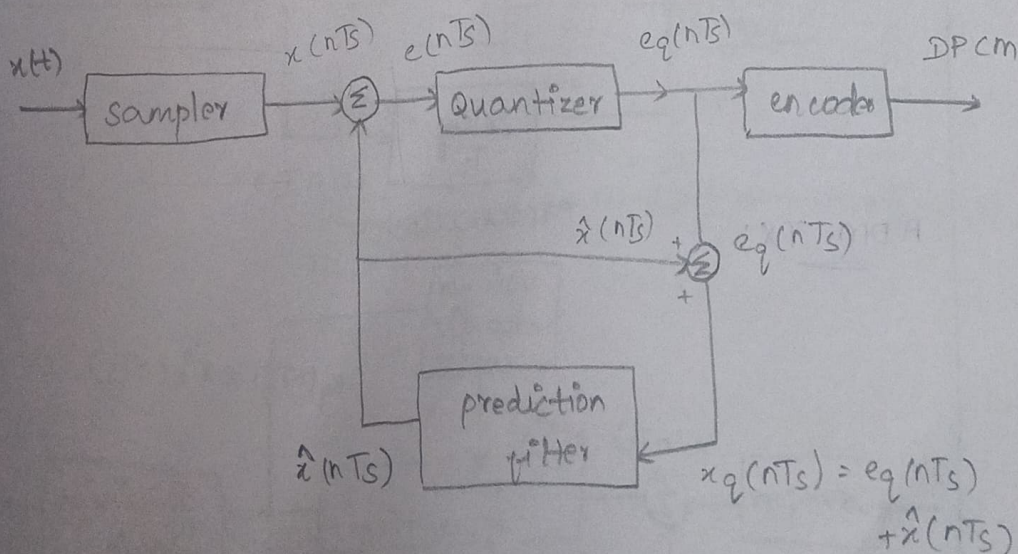
ADPCM - Tx:



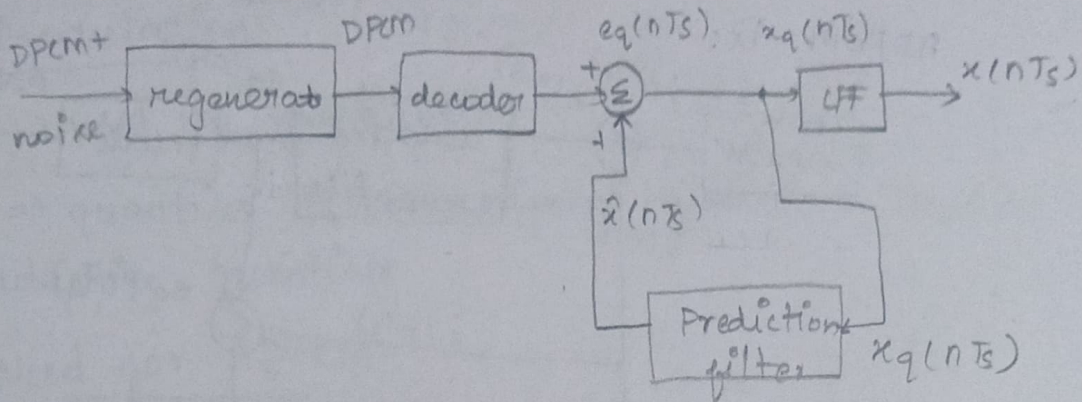
ADPCM - Rx:



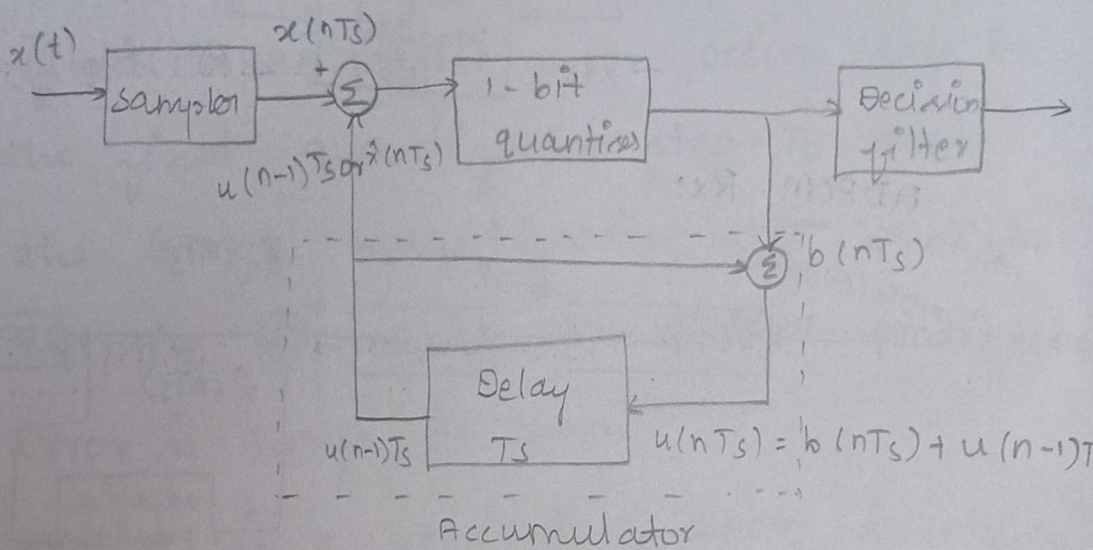
DPCM Tx:



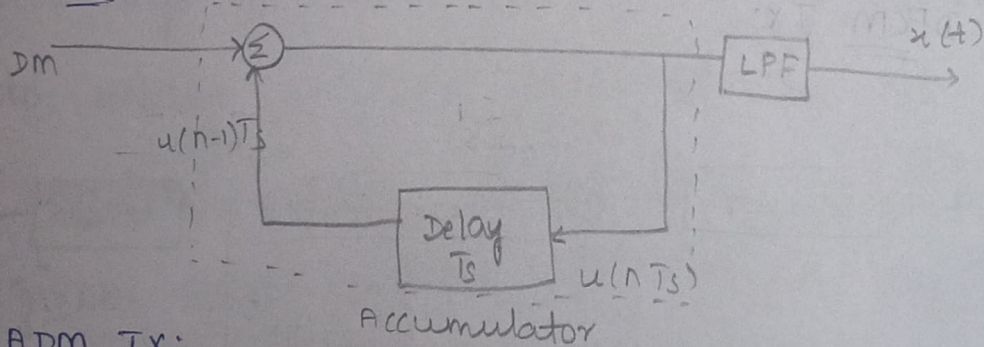
DPCM RX:



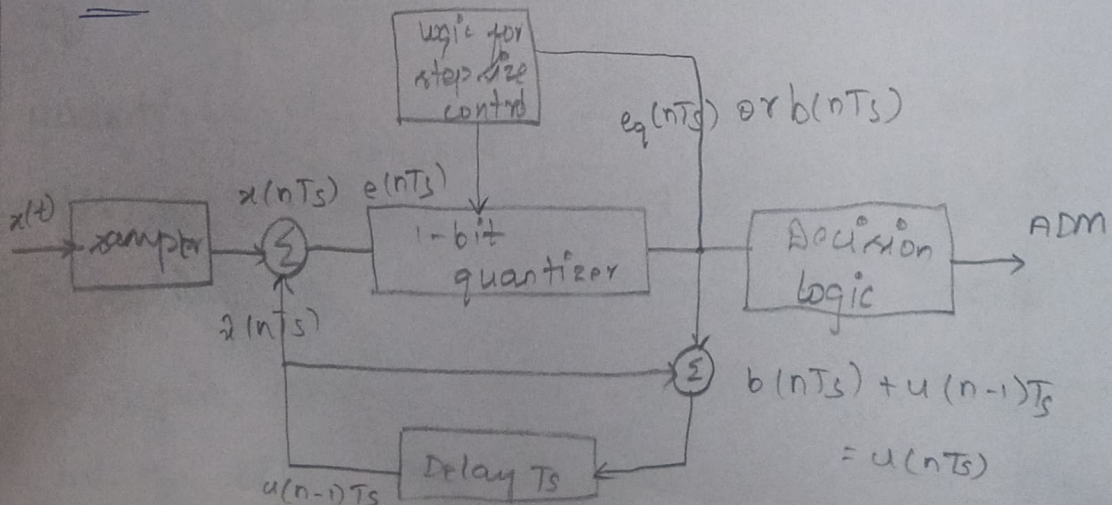
DM TX:

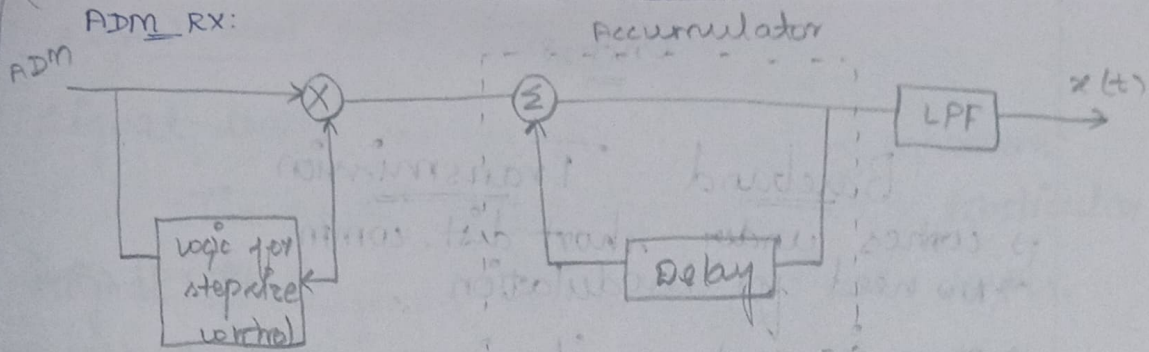


DM RX:

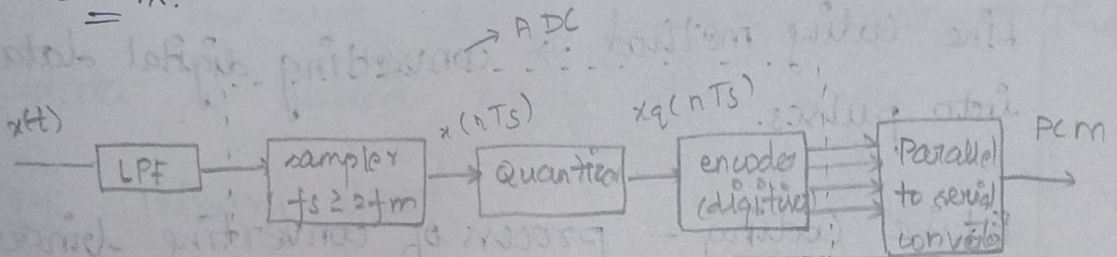


ADM TX:

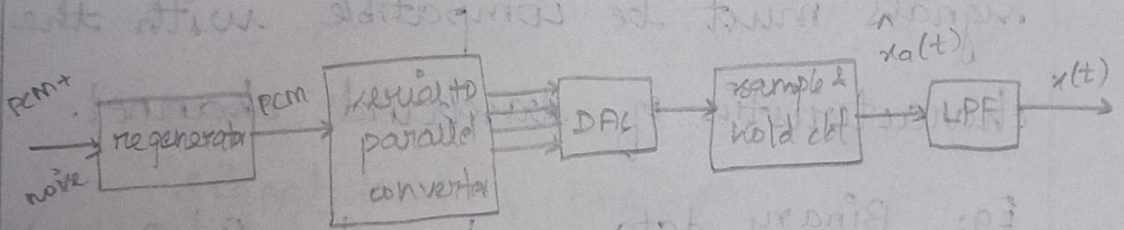




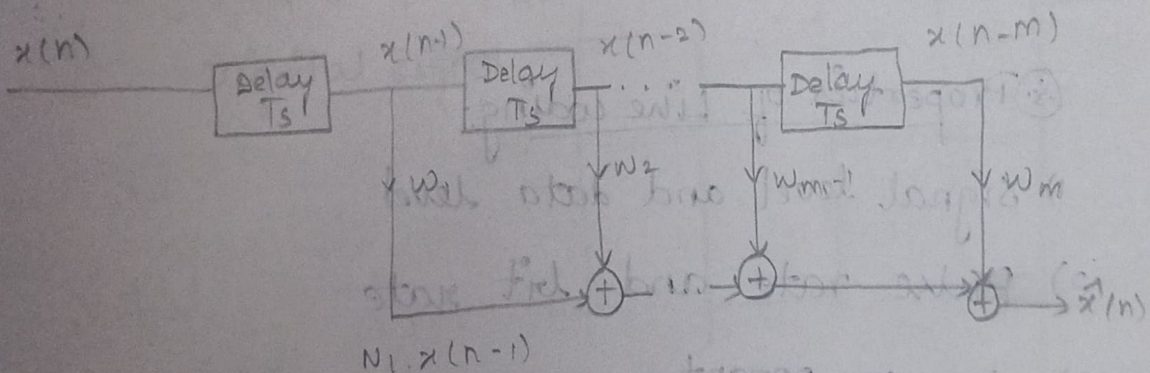
PCM TX:



PCM RX:



Prediction filter:



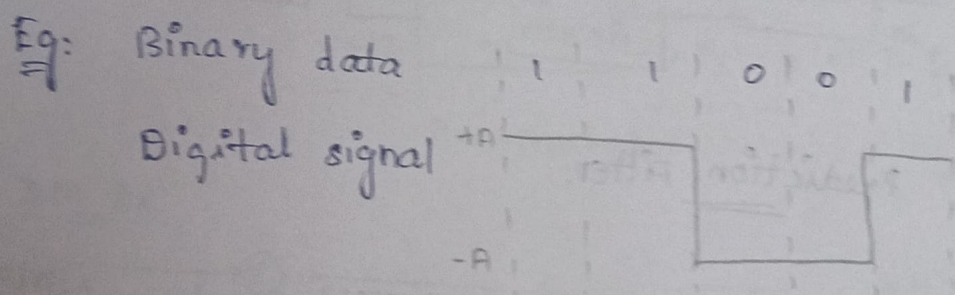
Unit-3

Baseband Transmission

- comes under short dist. comm.
- No need for modulation we are not sending modulated signal
- msg is sent as it is through the channel. instead we send pulses

Line coding method → converting digital data into pulses.

Line coding - process of converting binary data into a digital signal. These digital signals must be compatible with the characteristics of baseband channel.



① Properties of Line coding:

- 1) Signal level and data level
- 2) Pulse rate and bit rate
- 3) DC component
- 4) Self-synchronization
- 5) Transmission BW
- 6) Low probability of error
- 7) Transmission power efficiency
- 8) Noise and interference tolerance

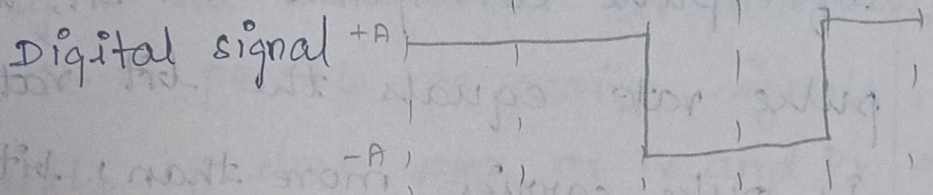
1) Signal level and data level:

Signal level: to represent
The no. of values allowed in a particular
signal is defined as no. of signal levels.

Data level:

The no. of values used to represent the
data is called no. of data levels.

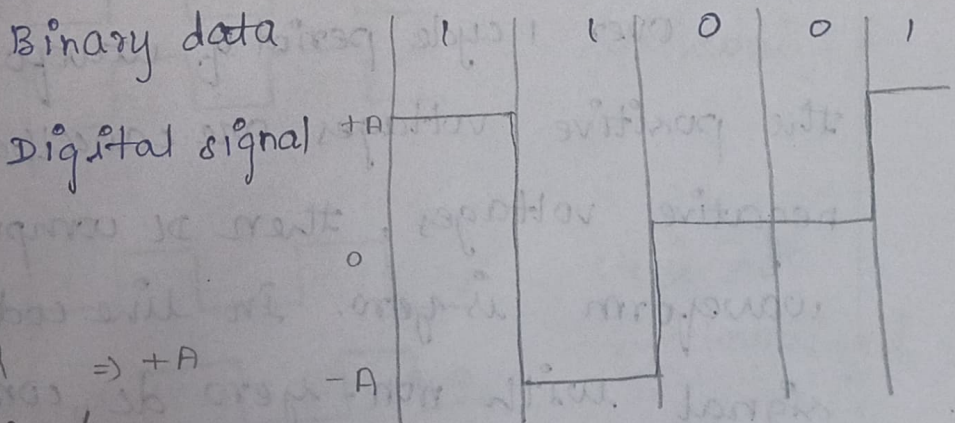
Eg-1: Binary data 1 1 1 0 0 1 1



Data level = 2 (0, 1)

Signal level = 2 ($+A, -A$)

Eg-2:



initial $\Rightarrow +A$

successive /

consecutive '1's

$\Rightarrow$ invert the polarity

0 $\Rightarrow$ zero level

Data level = 2 (0, 1)

Signal level = 3 ($A, 0, -A$)

In both examples, pulse rate = bit rate

2) Pulse rate and Bit rate:

Pulse rate:- is defined as no. of pulses transmitted per second. A pulse is defined as min. amount of time required to transmit a symbol.

Bit rate:- is defined as no. of bits transmitted per second.

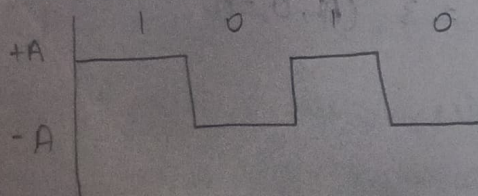
If 1 pulse corresponds to 1 bit, then pulse rate equals the bit rate.

If pulse carries more than 1 bit, then pulse rate is lower than the bit rate.

3) DC component:

Over 1 cycle period of waveform, if all the positive voltages are cancelled by negative voltages, then DC component of the waveform is zero. In line coding, the signal with non-zero DC component is treated as distorted one and it creates error in received signal.

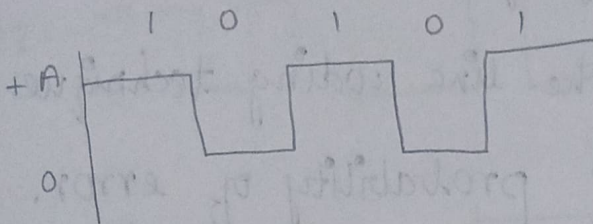
Eg: $1 \rightarrow +A$
 $0 \rightarrow -A$



DC comp = 0

1 $\rightarrow$ +A

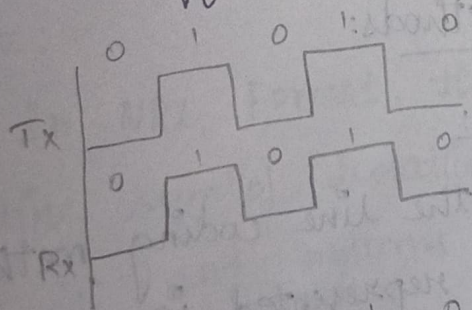
0 $\rightarrow$ 0



Average value = +A

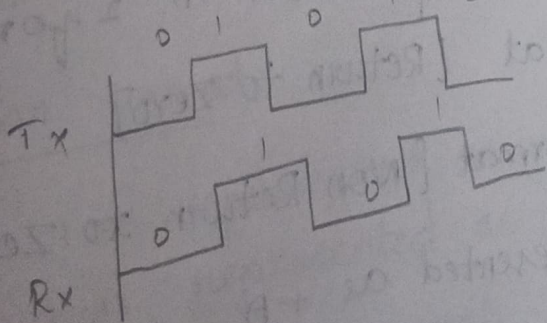
4) Self-synchronization:

If the receiver bit intervals corresponds exactly to the transmitter bit intervals, then it is possible to receive signals correctly. If clock freq. at Tx and Rx are different then the data at the Tx and Rx is different resulting in error.



clock is

synchronized



clk is not

synchronized

5) Transmission BW:

The line coding technique should have lesser bandwidth.

b) Probability of error:

The line coding technique should have low probability of error.

1) Transmission power efficiency:

The transmission power should have low power requirement for line coding technique.

2) Noise and interference tolerance:

The line coding method should have high noise tolerance.

Line Coding Methods:

1. Unipolar format:

→ It is one of the line coding method.

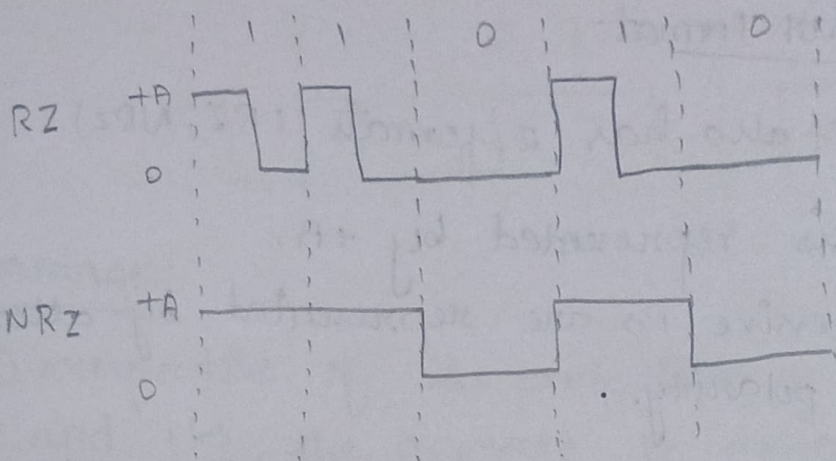
→ This has is represented in 2 formats

⇒ RZ format [Return to Zero]

⇒ NRZ format [Non Return to Zero]

⇒ 1 is represented as +A,

0 is represented as 0.



RZ $\Rightarrow$ after half pulse, it will return to zero.
(or at middle)

RI:

In RZ format, the pulses occupy a fraction of symbol duration i.e., the signal returns to zero at the middle of pulse duration.

NRZ:

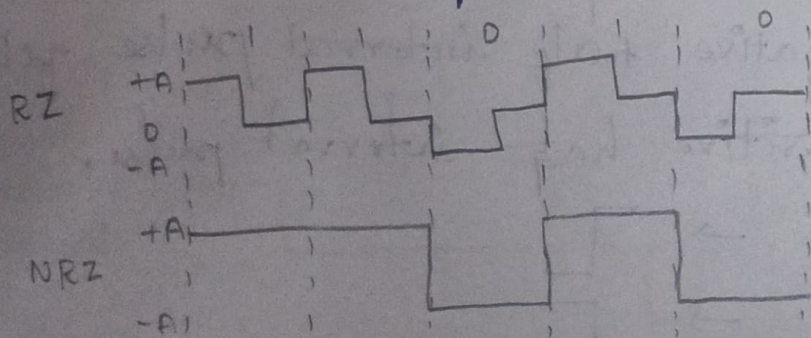
In NRZ Format, the pulses occupy entire symbol duration. It remains at $+A$ for '1' and returns to zero for '0'.

2. Polar Format:

$\Rightarrow$ 1 is represented as $+A$

0 is represented as $-A$

$\Rightarrow$ It has two formats (RZ, NRZ).



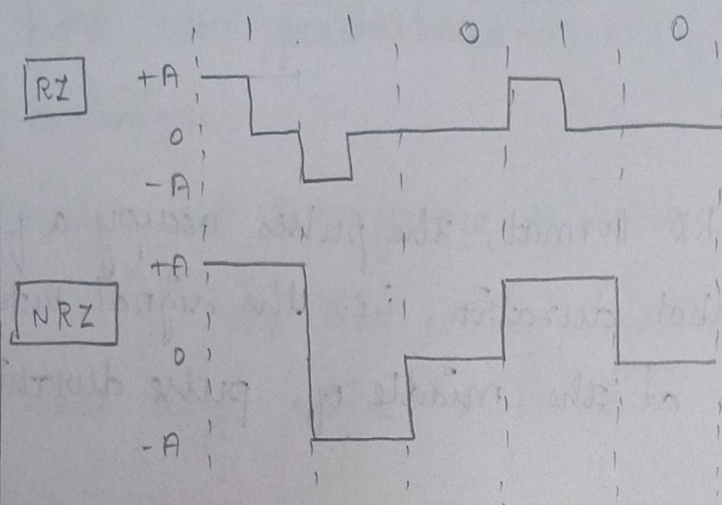
3. Bipolar format:

⇒ It also has 2 formats. (RZ, NRZ).

⇒ 1 is represented by $+A$.

⇒ Successive 1's are represented by alternating the polarity.

⇒ 0 is rep. as 0.



Advantages:

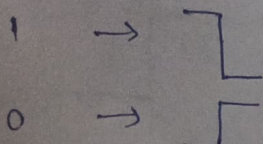
⇒ If there are even no. of 1's then the DC component of bipolar format is 0.

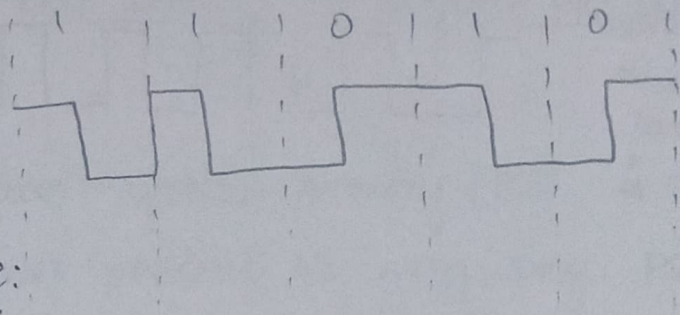
Split Phase Manchester:

⇒ It is a line coding technique.

⇒ Here, symbol 1 is represented by positive half interval pulse followed by negative half interval pulse. Symbol 0 is rep.

by negative half interval pulse followed by positive half interval pulse.





Advantage:

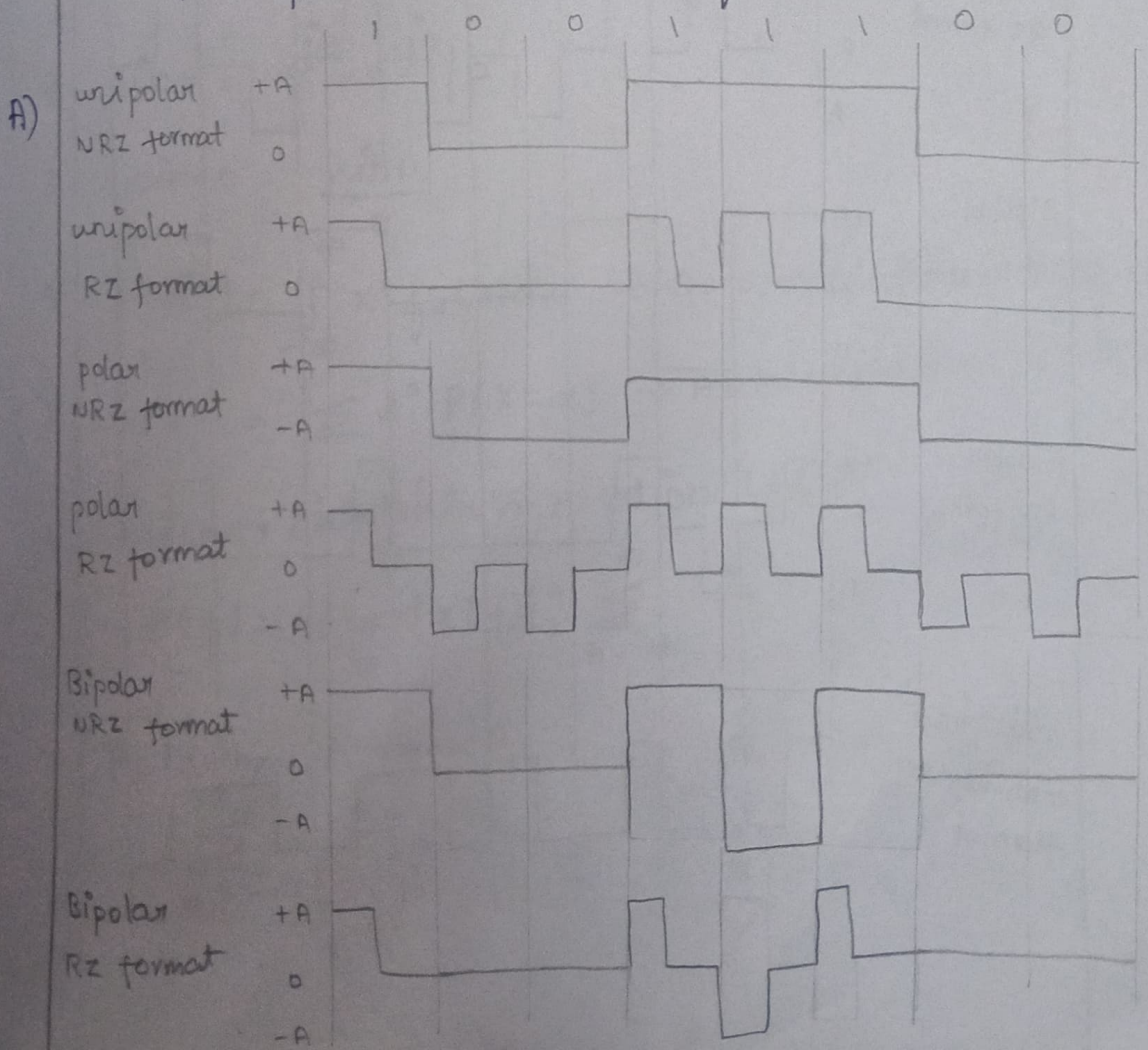
→ Irrespective of the probability of occurrence of 0's and 1's, the average DC component is zero. Hence power savings is more in this technique.

1. Represent the data using i) unipolar RZ, NRZ format

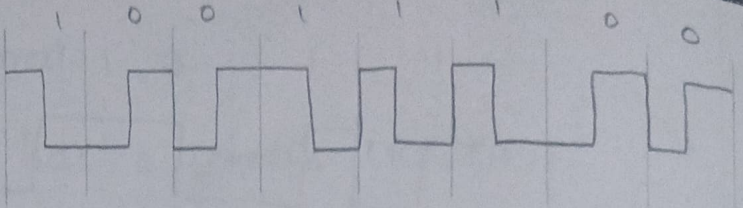
A) 1 0 0 1 1 0 0 B) 1 1 1 0 0 1 0 1

ii) Polar RZ, NRZ format iii) Bipolar RZ, NRZ Format

iv) split phase manchester format.

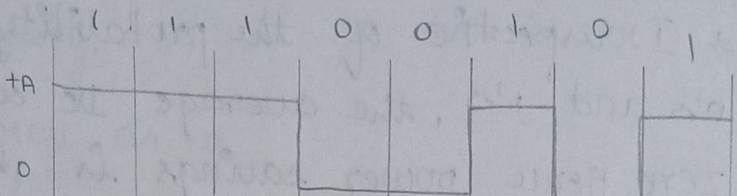


split phase
manchester
format

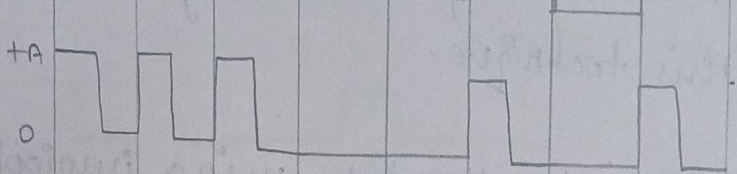


B)

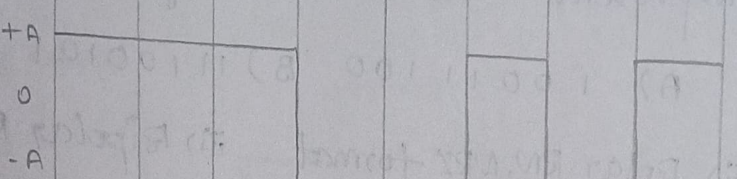
unipolar
NRZ format



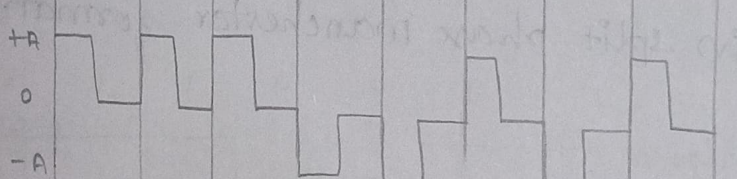
unipolar
RZ format



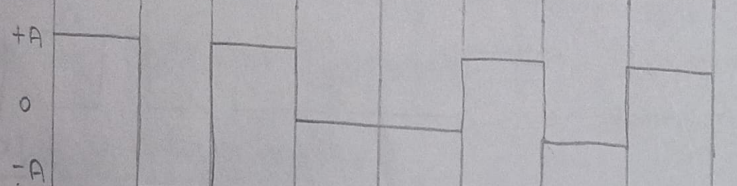
polar
NRZ format



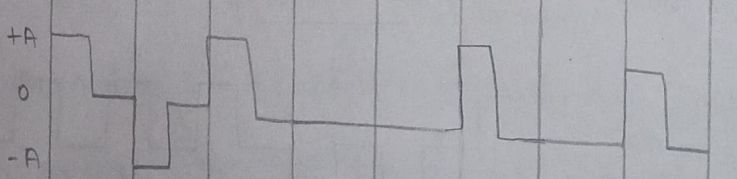
polar
RZ format



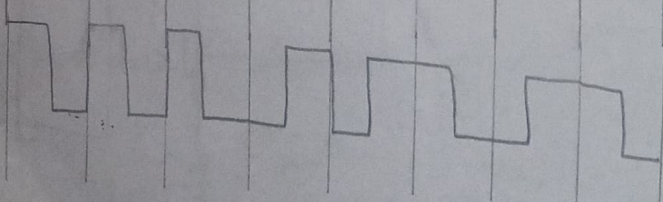
Bipolar
NRZ format



Bipolar
RZ format



split phase
manchester
format



Power Spectral Density of unipolar NRZ Line Coding scheme:

A power spectral density [PSD] is a measure of signal power content vs frequency. PSD is typically used to characterize a broad band random signal.

Auto-correlation - represents the degree of similarity b/w a given time series and the lagged version of itself over successive time intervals.

Note:

1. $X(f) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$

2. $\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2}$

3. $\text{sinc} \theta = \frac{\sin \theta}{\theta}$

4. $E(X^2) = \int_{-\infty}^{\infty} x^2 f_x(x) dx \rightarrow \text{Autocorrelation C.T domain}$

5. $E(X^2) = \sum x^2 P(X=x) \rightarrow \text{Autocorrelation D.T domain}$

6. Wiener khintchin's relation (PSD):

$$P(f) = \frac{1}{T_b} |X(f)|^2 = \sum_{n=-\infty}^{\infty} R_A(n) e^{-j2\pi f n T_b}$$

7. Poisson Formula:

$$\sum_{n=-\infty}^{\infty} e^{-j2\pi f n T_b} = \frac{1}{T_b} \sum_{m=-\infty}^{\infty} \delta(f - \frac{m}{T_b})$$

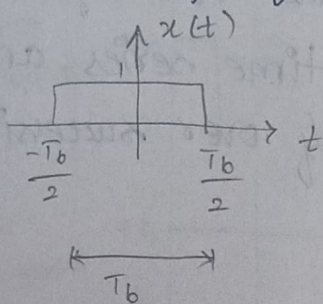
Step 1: Find FT of NRZ pulse.

Step 2: Find auto-correlation $R_A(n)$ of NRZ pulse.

Step 3: calculate PSD based on $x(f)$ and $R_A(n)$

PSD of unipolar NRZ pulse:

1. Find FT $X(f)$ of unipolar NRZ pulse.



$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$
$$= \int_{-T_b/2}^{T_b/2} (1) e^{-j2\pi f t} dt$$

$$= \left[\frac{e^{-j2\pi f t}}{-j2\pi f} \right]_{-T_b/2}^{T_b/2}$$

$$= \frac{e^{-j2\pi f \frac{T_b}{2}} - e^{j2\pi f \frac{T_b}{2}}}{-j2\pi f}$$

$$= \frac{1}{-j\pi f} \left[\frac{e^{j2\pi f \frac{T_b}{2}} - e^{-j2\pi f \frac{T_b}{2}}}{2j} \right]$$

$$= \frac{1}{\pi f} \sin \frac{2\pi f T_b}{2}$$

$$= \frac{1}{\pi f} \sin(\pi f T_b)$$

$$= \frac{T_b}{f} \left[\frac{\sin(\pi f T_b)}{\pi f T_b} \right]$$

$$X(f) = T_b \sin(\pi f T_b)$$

2. Find auto-correlation $R_A(n)$ of unipolar NRZ pulse.
 $\Rightarrow$ Assume the ^{probability} occurrence of 0's and 1's is equal.

$$P(A_c = 1) = P(A_c = 0) = \frac{1}{2}$$

$\Rightarrow$ unipolar : $1 \rightarrow +A$
 $0 \rightarrow 0$

$$R_A[n] = E[A_L A_{L-n}]$$

i) $n=0$:

$$R_A[0] = E[A_L A_L]$$

Bit	A_L	A_L	A_L^2	Prob
0	0	0	0	$\frac{1}{2}$
1	$+A$	$+A$	A^2	$\frac{1}{2}$

$$E[X^2] = \sum x^2 P(X=x)$$

$$R_A[0] = \sum A_L^2 P(X=x)$$

$$= 0\left(\frac{1}{2}\right) + A^2\left(\frac{1}{2}\right)$$

$$R_A[0] = \frac{A^2}{2}$$

ii) $n \neq 0$:

$$R_A(n) = E[A_L \cdot A_{L-n}]$$

Bit	A_L	A_{L-n}	$A_L \cdot A_{L-n}$	Prob
0 0	0	0	0	$\frac{1}{4}$
0 1	0	$+A$	0	$\frac{1}{4}$
1 0	$+A$	0	0	$\frac{1}{4}$
1 1	$+A$	$+A$	A^2	$\frac{1}{4}$

$$R_A[n] = 0\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right) + A^2\left(\frac{1}{4}\right) = \frac{A^2}{4}$$

$$\therefore R_A(n) = \begin{cases} \frac{A^2}{2}, & n=0 \\ \frac{A^2}{4}, & n \neq 0 \end{cases}$$

3. Calculate PSD of unipolar NRZ based on $x(t)$ and $R_A(n)$.

$$P(f) = \frac{1}{T_b} |x(t)|^2 \sum_{n=-\infty}^{\infty} R_A(n) e^{-j2\pi f n T_b}$$

$$= \frac{1}{T_b} |T_b \text{sinc}(\pi f T_b)|^2 \left[\frac{A^2}{2} (1) \right]$$

$$+ \frac{A^2}{4} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b}$$

$$= \frac{1}{T_b} T_b^2 \text{sinc}^2(\pi f T_b) \left[\frac{A^2}{2} + \frac{A^2}{4} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b} \right]$$

$$= T_b \text{sinc}^2(\pi f T_b) \left[\frac{A^2}{2} + \frac{A^2}{4} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b} \right]$$

$$= \frac{A^2}{2} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) \left[\left\{ \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} e^{-j2\pi f n T_b} \right\} + 1 - 1 \right]$$

$$= \frac{A^2}{2} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) \left[\left\{ \sum_{n=-\infty}^{\infty} e^{-j2\pi f n T_b} \right\} - 1 \right]$$

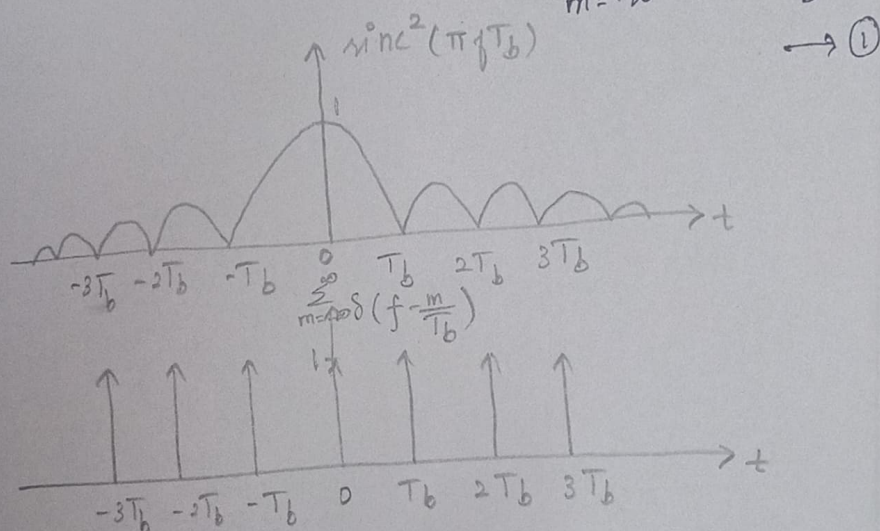
$$P(f) = \frac{A^2}{2} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) (-1) + \frac{A^2}{4} \left[\sum_{n=-\infty}^{\infty} e^{-j2\pi f n T_b} \right] (T_b \text{sinc}^2(\pi f T_b))$$

Apply Poisson's formula.

$$= \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b)$$

$$\left[\frac{1}{T_b} \sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T_b}\right) \right]$$

$$P(f) = \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} \sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T_b}\right)$$



$$\textcircled{1} \Rightarrow \boxed{P(f) = \frac{A^2}{4} T_b \text{sinc}^2(\pi f T_b) + \frac{A^2}{4} \delta(f)} \rightarrow \textcircled{2}$$

Eqn. ② represents the PSD of unipolar NRZ format. The second term $\frac{A^2}{4} \delta(f)$ is the DC component present in the unipolar NRZ format.